PART 2 | CONSTRAINT CARDS

Constraint cards

Each card defines a surprise condition, operating rules and an observable pass condition. Organisers should use the same challenge mechanics across tracks while substituting the track-specific data and workflow.

CONSTRAINT CARD C01

The World Went Offline

SURPRISE CONDITION

All external internet connectivity and third-party API access become unavailable during the challenge. The application must continue its essential user journey using information and components available within the local environment.

RULES Teams must support local access to essential saved information; safely queue eligible new actions; prevent data loss and duplicate operations; and synchronise pending changes when connectivity returns. External hosted AI/LLM inference may not be used by the application during this test.

WHAT TEAMS MUST DEMONSTRATE Demonstrate which functions remain available offline, how the application communicates reduced functionality, how pending actions are queued and how reconnection is handled.

PASS CONDITION

The core user journey remains operational without internet, and pending changes synchronise after reconnection without losing or duplicating operations.

CONSTRAINT CARD C02

The Truth Is Conflicting

SURPRISE CONDITION

Two supposedly reliable sources provide conflicting information, and at least one record is stale. The system must not silently treat the newest-arriving record as the truth.

RULES Teams must inspect timestamps and source information; apply a justified source-priority or freshness rule where appropriate; identify unresolved inconsistencies; and request verification where the correct value cannot be determined safely. Unverified information must not silently become authoritative.

WHAT TEAMS MUST DEMONSTRATE Demonstrate conflict detection, the rule used to compare records, the selected or unresolved value, and an explanation of what the system trusts and why.

PASS CONDITION

The system identifies the conflict and explains which information it trusts, why it trusts it, and what remains uncertain.

CONSTRAINT CARD C03

The API Has Changed Its Language

SURPRISE CONDITION

A previously working API suddenly changes its response structure. A field may be renamed, moved into a nested object, change type or become optional.

RULES The original integration must fail without adaptation. Teams should isolate external data contracts behind an adapter or normalisation layer; validate incoming responses; and prevent API-specific fields from spreading throughout the application. Invalid or unknown responses must be handled gracefully.

WHAT TEAMS MUST DEMONSTRATE Demonstrate how the changed response is mapped into the application's internal data model, how unexpected fields are handled and why the rest of the application does not require a rewrite.

PASS CONDITION

The application adapts to the new contract without requiring an entire rewrite, and invalid responses do not crash the core workflow.

CONSTRAINT CARD C04

Garbage In, Garbage Everywhere

SURPRISE CONDITION

The application receives unexpected, malformed or malicious inputs. The injected dataset may contain missing fields, duplicates, invalid timestamps, negative or unrealistic values, unexpected Unicode, malformed JSON/CSV or unexpected data types.

RULES The application must validate data before processing; reject or quarantine invalid records; preserve valid records and continue processing; return useful errors; and avoid injection vulnerabilities. Teams must not silently coerce unsafe values into plausible-looking records.

WHAT TEAMS MUST DEMONSTRATE Demonstrate validation at the input boundary, safe handling of malformed records, a useful error or audit entry and continued processing of valid data.

PASS CONDITION

The application processes valid records, safely rejects or quarantines invalid ones and remains available without corrupting stored data.

CONSTRAINT CARD C05

Three Times the Crowd, Half the Resources

SURPRISE CONDITION

Demand suddenly increases to three times the original level while the available resource pool is reduced by 50%.

RULES Teams must recompute allocation; respect existing commitments or reservations according to stated rules; keep priority rules transparent; never allocate more resources than exist; and provide alternatives or wait-list states for requests that cannot be fulfilled.

WHAT TEAMS MUST DEMONSTRATE Demonstrate a feasible reallocation, capacity totals, the priority rule applied and what happens to users who cannot be served immediately.

PASS CONDITION

The solution produces a feasible allocation, never exceeds available capacity and explains who receives resources, who does not and why.

CONSTRAINT CARD C06

Accessibility Is No Longer Optional

SURPRISE CONDITION

A selected user can no longer use the application's primary interaction method. The test may require keyboard-only use, screen-reader-compatible structure, no reliance on colour, no audio-only instructions or an alternative to complex gestures.

RULES The primary workflow must remain accessible; important information cannot rely exclusively on colour, icons or sound; navigation must support an alternative interaction method; and errors and confirmations must remain understandable.

WHAT TEAMS MUST DEMONSTRATE Complete the selected critical workflow using the supplied accessibility condition and show that essential content and status remain perceivable and operable.

PASS CONDITION

The selected workflow can be completed using the specified accessibility condition without losing access to essential information or actions.

CONSTRAINT CARD C07

Your Data Cannot Leave This Device

SURPRISE CONDITION

A new privacy restriction is imposed. Specified personal or sensitive records cannot be transmitted to external services or processed by a cloud-hosted AI service.

RULES Teams must keep protected information within the approved local environment; use local processing or synthetic/anonymised data where necessary; prevent sensitive content from leaking into logs; and preserve the essential workflow within the new boundary.

WHAT TEAMS MUST DEMONSTRATE Show the data-flow boundary, identify which components can access protected data, demonstrate that prohibited outbound transmission is blocked and show safe logging.

PASS CONDITION

The system works under the new privacy boundary, with no prohibited data transmission and no sensitive information exposed through logs or error messages.

CONSTRAINT CARD C08

The Language Changed Overnight

SURPRISE CONDITION

The application's primary language changes, but the supplied translation resources are incomplete. Organisers specify the target language and provide or identify the required critical terminology.

RULES Teams must support the second language; provide a safe fallback for missing non-critical strings; preserve the meaning of critical instructions; ensure text expansion does not break the interface; and keep errors understandable. Do not invent translations for critical safety or operational instructions.

WHAT TEAMS MUST DEMONSTRATE Demonstrate a complete primary workflow in the target language, sensible handling of missing strings and usable layouts for longer text.

PASS CONDITION

Users can complete the primary workflow in the new language without broken screens, missing critical instructions or misleading translations.

CONSTRAINT CARD C09

Two People, One Resource

SURPRISE CONDITION

Two or more users attempt to claim or modify the same resource at exactly the same time. The organiser may simulate near-simultaneous requests, retries or duplicate submissions.

RULES Teams must handle concurrent requests correctly; prevent double bookings and duplicate allocations; preserve database consistency; make retries idempotent where appropriate; and return accurate success or failure responses.

WHAT TEAMS MUST DEMONSTRATE Demonstrate concurrent requests against a scarce resource and show the resulting state in the database or audit log.

PASS CONDITION

Under concurrent requests, the application never allocates more resources than actually exist and all users receive consistent outcomes.

CONSTRAINT CARD C10

The Sensors Are Lying

SURPRISE CONDITION

Some incoming sensor or telemetry readings become unreliable. The injected feed may include missing readings, sudden spikes, repeated values, unusual fluctuations, delayed observations or conflicting readings from different sensors.

RULES Teams must detect suspicious readings; distinguish a possible sensor fault from a genuine operational event; avoid acting blindly on one faulty reading; and provide a safe fallback, cross-check or human verification step.

WHAT TEAMS MUST DEMONSTRATE Demonstrate a fault-detection rule, how the system marks uncertain readings and what it does instead of blindly triggering an action.

PASS CONDITION

The solution detects the injected fault and avoids making an unsafe or unjustified automated decision.

CONSTRAINT CARD C11

The Rules Changed Midway

SURPRISE CONDITION

The organisers issue a new operational policy after the solution has already generated its initial result. The updated policy changes a priority, eligibility, safety or operating rule.

RULES The system must recognise the policy update; re-evaluate affected recommendations or allocations; keep previous records consistent; record the policy version or change; and require appropriate confirmation before irreversible actions.

WHAT TEAMS MUST DEMONSTRATE Demonstrate the before-and-after result, show which decisions were recalculated and identify any actions that require approval rather than silent reversal.

PASS CONDITION

The system adapts to the new policy without corrupting previous records or continuing to apply obsolete rules.

CONSTRAINT CARD C12

The Missing Piece of Information

SURPRISE CONDITION

The system has enough information to generate possible actions but not enough information to safely choose one. The scenario intentionally omits one decision-critical fact.

RULES Teams must identify exactly what is missing; must not invent the missing information; must ask a relevant clarification or escalate to an authorised human; must avoid irreversible actions until necessary details are confirmed; and should explain why the system cannot safely proceed.

WHAT TEAMS MUST DEMONSTRATE Demonstrate the missing-field or uncertainty check, the clarification or escalation request and the blocked or deferred action.

PASS CONDITION

Instead of guessing, the application identifies the missing information and makes the correct request for clarification or human review.